

Lecture 12: Solving Linear Programming Problems II

MARCH 2

Lecturer: Mitchell Scott

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12.0.1 Extreme Points

Definition 12.1. A point $\mathbf{x} \in \mathbb{R}^n$ is a convex combination of the points $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_r$ in $\mathbb{R}^n$ if for some real numbers $c_1, c_2, \dots, c_r$ satisfy

$$\sum_{i=1}^r c_i = 1 \quad \text{and} \quad c_i > 0, 1 \leq i \leq r$$

we have

$$\mathbf{x} = \sum_{i=1}^r c_i \mathbf{x}_i.$$

Definition 12.2. A point $x \in C$ is an extreme point if it cannot be written as a strict convex combination of two distinct points in C .

Remark 12.3. We already saw this in two dimensions! Consider $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^2$. If we have $c_1 = p$ and $c_2 = 1 - p$, then we have $\mathbf{x} = \sum_{i=1}^2 c_i \mathbf{x}_i = p\mathbf{x}_1 + (1 - p)\mathbf{x}_2$. These are all the interior points.

For the extreme points, we have the end points. The end points require $p = 0, 1$ but that means not all of the $c_i > 0$.

Definition 12.4. A bounded convex set is one that can be enclosed in a rectangle in $\mathbb{R}^n$. An unbounded convex set cannot be so enclosed. The convex sets illustrated in Figure 1.9 are bounded ; those in Figure 1.10 are unbounded .

Theorem 12.5 (Fundamental Theorem of LP). Let S be the set of feasible solutions to a general linear programming problem.

1. If S is nonempty and bounded, then an optimal solution to the problem exists and occurs at an extreme point.
2. If S is nonempty and not bounded and if an optimal solution to the problem exists, then an optimal solution occurs at an extreme point.
3. If an optimal solution to the problem does not exist, then either S is empty or S is unbounded.

Proposition 12.6. The feasible region of a linear program is convex.

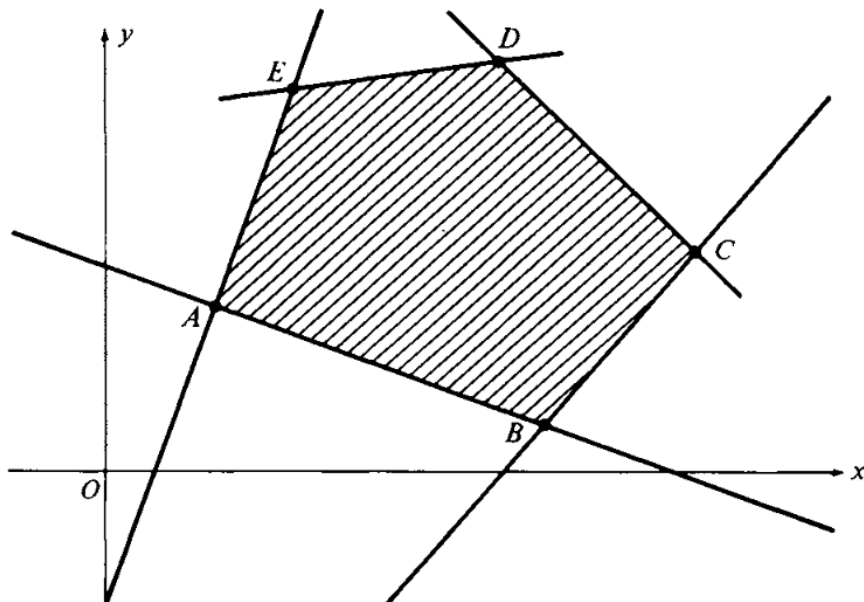


Figure 12.1:

Example 12.7. Consider the linear programming problem

maximize $z = 120x + 100y$ subject to

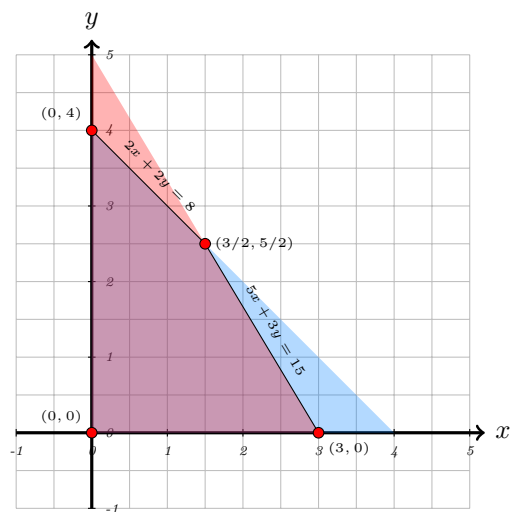
$$2x + 2y \leq 8$$

$$5x + 3y \leq 15$$

$$x > 0$$

$$y > 0$$

The convex set of all feasible solutions are shown below:



Extreme point (x, y)	Value of objective function $z = 120x + 100y$
(0, 0)	0
(3, 0)	360
(0, 4)	400
$(\frac{3}{2}, \frac{5}{2})$	430

We see that the extreme points of the convex set S are in red above: $(0, 0)$, $(3, 0)$, $(0, 4)$, and $(\frac{3}{2}, \frac{5}{2})$. Since S is nonempty and bounded (check these), the objective function attains its maximum at an extreme point, by the above theorem. This means we only have 4 points to check and compare the values.

This means that the maximum value of z occurs at $(\frac{3}{2}, \frac{5}{2})$, so the lumber mill should saw 1500 board of finish-grade lumber and 2500 feet of construction grade lumber per day to earn a maximum profit of \$430 a day.

Example 12.8. Consider the linear programming problem

maximize $z = 2x + 5y$ subject to

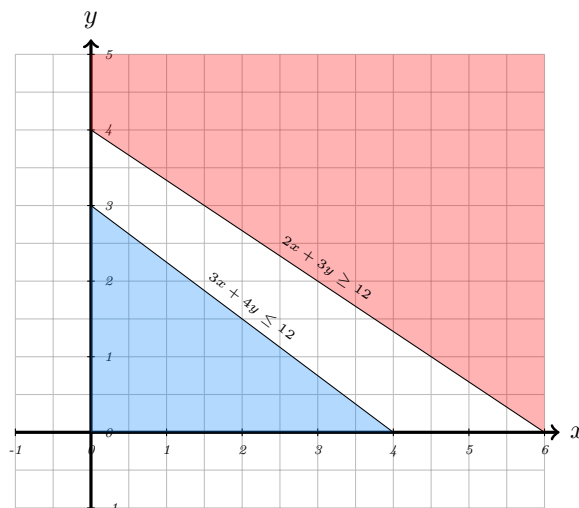
$$2x + 3y \geq 12$$

$$3x + 4y \leq 12$$

$$x > 0$$

$$y > 0$$

The convex set of all feasible solutions are shown below:

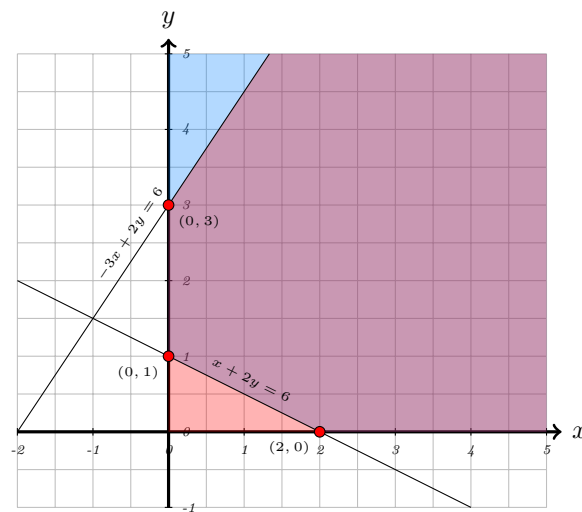


There are no such points that are in both of those regions, (no purple). Since there are no feasible solutions, there are no optimal solutions.

Example 12.9. Consider the linear programming problem

$$\begin{aligned} \text{minimize } z &= 2x + 5y \text{ subject to} \\ -3x + 2y &\leq 6 \\ x + 2y &\geq 2 \\ x &> 0 \\ y &> 0 \end{aligned}$$

The convex set of all feasible solutions are shown below:



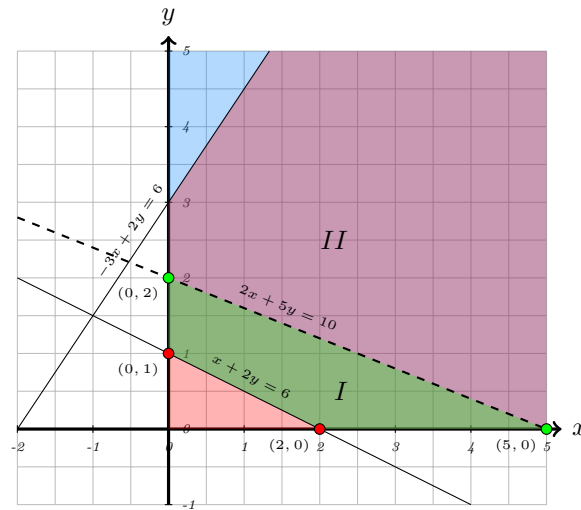
There are many points that are in the feasible solution, so even though we have an unbounded domain, since we know that there is an optimal solution, then it must occur at one of the extreme points, which are highlighted in red. We look at the table below:

Extreme point (x, y)	Value of objective function $z = 2x + 5y$
(0, 3)	15
(0, 1)	5
(2, 0)	4

We see that (2, 0) is the optimal solution, which is the same that we found last week.

There might be another way to solve this, which first finds a bounded set, which means that the solution occurs at an extreme point.

First, we consider two regions Region II which is the feasible region above $2x + 5y = 10$, and Region I, which is the feasible region below $2x + 5y = 10$. Since we are trying to minimize this function, we can safely ignore anything that is above $k = 10$, or the entirety of Region II. This now means our region of interest can be viewed as the bounded region in green, or Region I.



This means that we now have a bounded, nonempty solution region, which means that there must be an optimal solution. We can then fill out the table below:

Extreme point (x, y)	Value of objective function $z = 2x + 5y$
(0, 2)	10
(5, 0)	10
(0, 1)	5
(2, 0)	4

Using this method, we also see that we have the same optimal solution, which is $(x, y) = (2, 0)$.

Example 12.10. Consider the linear programming problem

maximize $z = 2x + 3y$ subject to

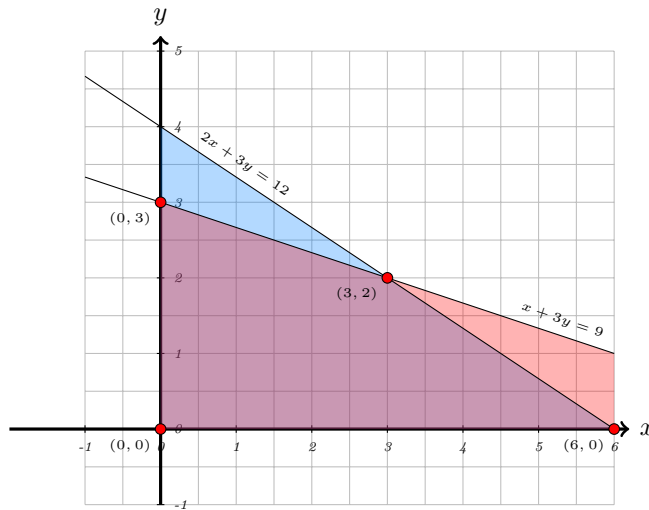
$$x + 3y \leq 9$$

$$2x + 3y \leq 12$$

$$x > 0$$

$$y > 0$$

The convex set of all feasible solutions are shown below:



This means that we now have a bounded, nonempty solution region, which means that there must be an optimal solution. We can then fill out the table below:

Extreme point (x, y)	Value of objective function $z = 2x + 3y$
(0, 0)	0
(0, 3)	9
(6, 0)	12
(3, 2)	12

Using this method, we see that both extreme points of (6, 0) and (3, 2) give the maximal value of 12. We also know that the line segment joining these two points are

$$\begin{aligned}
 (x, y) &= p(6, 0) + (1 - p)(3, 2) \\
 &= (6p, 0) + (3 - 3p, 2 - 2p) \\
 &= (3 + 3p, 2 - 2p) \quad \text{for } 0 \leq p \leq 1
 \end{aligned}$$

For any point (x, y) on this line segment, we have

$$\begin{aligned}
 z &= 2x + 3y = 2(3 + 3p) + 3(2 - 2p) \\
 &= 6 + 6p + 6 - 6p \\
 &= 12.
 \end{aligned}$$

This means that any point on this line segment is an optimal solution.